



```

1 . do "C:\Users\fahim\AppData\Local\Temp\STD962c_000000.tmp"

2 . clear all

3 . import excel "C:\Users\fahim\Downloads\PV Loop Statistics-Raw Data.xlsx", sheet("FINAL-Cumulative Sheet") first
  (24 vars, 162 obs)

4 . destring StrokeWork StrokeVolume EjectionFraction HeartRate CardiacOutput Elastance ESV ESP EDV EDP, replace fo
  StrokeWork: all characters numeric; replaced as double
  (14 missing values generated)
  StrokeVolume: all characters numeric; replaced as double
  (14 missing values generated)
  EjectionFraction: all characters numeric; replaced as double
  (14 missing values generated)
  HeartRate: all characters numeric; replaced as double
  (14 missing values generated)
  CardiacOutput: all characters numeric; replaced as double
  (14 missing values generated)
  Elastance: all characters numeric; replaced as double
  (14 missing values generated)
  ESV: all characters numeric; replaced as double
  (14 missing values generated)
  ESP: all characters numeric; replaced as double
  (14 missing values generated)
  EDV: all characters numeric; replaced as double
  (14 missing values generated)
  EDP: all characters numeric; replaced as double
  (14 missing values generated)

5 . by Group CavalOcclusion, sort : tabstat StrokeWork StrokeVolume EjectionFraction HeartRate CardiacOutput Elasta

```

-> Group = A, CavalOcclusion = Before Occlusion

Summary statistics: Mean, SD
Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
0	3300.779 651.2895	44.59288 10.79583	42.19551 10.94459	91.2649 19.46178	4039.32 1090.473	1.797607 .5355723	63.26929 17.6923	74.92483 7.462314	107.8622 13.14432	6.7157 2.27
30	1451.49 636.2868	38.60353 10.87617	59.00587 10.59054	127.6794 34.60548	5118.434 2538.332	1.055738 .3559056	26.58466 6.365856	37.34864 7.708619	65.18819 11.14129	4.3662 1.9977
74	4226.418 801.835	55.86774 9.228837	53.76813 5.918314	136.489 19.86399	7657.647 1851.621	1.359533 .4652475	48.40281 5.888	71.56783 15.82517	104.2706 9.313728	7.1947 2.0421
Total	2992.896 1357.197	46.35472 12.1804	51.6565 11.43459	118.4778 31.39136	5605.134 2385.157	1.404293 .5324169	46.08559 18.82843	61.28044 20.31532	92.4403 22.54543	6.0922 2.3528

-> Group = A, CavalOcclusion = During Occlusion

Summary statistics: Mean, SD
Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
0	1973.599 681.2918	35.45588 10.08143	47.58784 13.67715	92.34935 19.5722	3273.551 1060.509	1.721995 .4058325	42.8824 20.40156	56.74166 5.70648	78.33828 22.24007	4.8062 3.5956
30	742.4851 344.326	25.26234 7.575864	49.52993 8.489366	123.9247 29.58617	3281.553 1735.286	1.383121 .387563	25.455 5.805319	32.15589 5.366888	50.71734 10.49949	3.756 2.3619
74	2107.82 653.1548	41.77589 4.427638	54.42331 6.520457	136.0648 21.64432	5675.963 851.4298	1.367769 .3996595	34.62801 8.309353	53.43872 10.50383	76.40391 8.801056	5.6291 .83877
Total	1553.875 837.5009	33.60831 10.13401	50.45221 9.673483	117.8512 29.34238	4027.305 1677.746	1.484222 .4050145	33.76763 14.00781	46.48983 13.45437	67.37593 19.17108	4.6696 2.4780

-> Group = A, CavalOcclusion = Post Occlusion

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
0	3190.451 619.8217	43.68592 11.79505	41.9192 12.1123	91.93437 20.15378	3994.407 1237.265	1.858181 .7606676	63.03791 19.20122	72.74458 7.989453	106.7238 13.70272	6.3619 2.70
30	1686.309 718.3566	41.13431 12.21827	61.68288 9.705406	119.4622 26.02211	5070.637 2367.748	1.034409 .3237531	25.16746 5.530865	39.02505 7.675295	66.30178 13.38666	3.1485 2.3616
74	4427.448 1195.32	53.1681 10.42266	51.75559 7.18528	133.2405 21.58967	7090.892 1767.936	1.473601 .406528	50.1649 9.350413	74.80478 13.81757	103.333 12.64609	4.9063 2.2055
Total	3101.403 1419.567	45.99611 12.04813	51.78589 12.44865	114.879 27.71157	5385.312 2182.404	1.455397 .6079021	46.12342 20.12176	62.19147 19.41984	92.11953 22.58052	4.8056 2.6555

-> Group = B, CavalOcclusion = Before Occlusion

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
0	3647.849 1097.671	50.29586 20.69624	41.83666 9.554379	89.77123 14.62771	4764.248 2877.729	1.646709 .5661481	68.37402 15.7128	72.08196 3.667389	118.6699 27.02185	6.5888 3.5318
30	1823.478 460.3094	46.24147 14.04821	60.24399 8.412367	117.4871 23.27967	5541.693 2294.64	.9579525 .4342675	30.87836 10.95186	39.29132 7.843378	77.11984 21.30004	1.728 2.6064
74	9327.834 3237.75	59.3944 33.20702	45.15966 13.39959	163.4134 16.45323	9710.942 5632.721	3.866608 2.024224	68.86765 21.0455	173.7537 18.8724	128.262 43.24913	6.4378 1.5768
Total	5115.97 3814.943	52.31464 23.65669	48.4234 12.89404	123.9143 36.36043	6738.8 4363.672	2.227627 1.750841	57.52011 23.66056	98.3218 60.09792	109.8347 37.6457	5.1066 3.3805

-> Group = B, CavalOcclusion = During Occlusion

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
0	1763.866 636.3753	36.09121 11.6687	45.16982 8.679726	92.86425 19.46652	3480.621 1726.925	1.688282 .5784203	44.30814 7.060807	55.29482 8.756373	80.39935 11.83705	4.8351 3.6624
30	414.2451 151.2129	22.77059 5.117655	39.46394 2.790275	113.2277 19.14748	2625.836 936.9619	1.407614 .7822457	35.61081 12.24025	28.55139 7.454866	58.3814 17.29283	2.5534 3.5787
74	5680.823 4349.506	49.15152 30.91598	51.27918 14.91686	154.8887 34.13659	7830.465 5405.084	2.687129 1.063257	43.23705 10.84602	108.8578 42.20607	92.38857 33.72687	4.5747 3.985
Total	2873.575 3312.915	37.90119 21.29585	46.12905 10.93238	119.3794 37.13344	4850.969 3961.688	1.98487 .9369354	42.06189 9.533324	68.69373 40.75925	79.96308 24.83652	4.253 3.5902

-> Group = B, CavalOcclusion = Post Occlusion

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
0	3447.825 1081.362	51.69108 18.66839	45.61358 7.388418	88.89385 13.51274	4795.332 2645.618	1.429987 .4296798	61.72396 17.02055	66.23691 6.94997	113.415 29.59697	6.1242 2.7956
30	1908.116 423.9303	47.84624 14.24922	62.51347 5.449292	111.2249 22.28429	5389.5 2188.938	.9150061 .4208577	29.02344 9.645624	39.07541 8.645999	76.86969 22.0007	.71302 2.9143
74	9282.412 2973.628	61.41514 32.38962	49.22355 11.98031	161.5489 15.37274	9846.075 5194.137	3.252511 1.648022	61.05247 15.81818	152.7459 13.82227	122.4676 39.4775	5.2076 1.5996
Total	5054.235 3733.434	53.99227 22.84647	51.85824 11.04722	121.1048 35.88926	6752.702 4173.957	1.921766 1.422052	51.86917 20.57009	88.7808 50.89663	105.8614 35.62648	4.2092 3.3065

-> Group = E, CavalOcclusion = Before Occlusion

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
0	3201.441 743.9599	42.94326 14.60021	44.46814 12.88167	88.48033 18.87573	3821.76 1487.768	2.015413 .747457	55.26398 21.34669	76.10443 5.082295	98.20724 22.80748	7.0217 3.377
30	1840.657 342.1293	39.2323 5.807126	62.15725 7.988782	135.4013 31.0769	5382.582 1848.658	1.230866 .278158	24.80008 8.723949	46.69055 6.449735	64.03238 11.64702	5.0446 5.2032
74	4771.319 870.8288	34.15652 .9270961	38.97845 6.076406	135.0494 27.77898	4598.715 883.7044	4.071784 .6934906	55.64809 15.95962	134.6173 22.42624	89.80462 15.69091	7.4633 4.0435
Total	3182.893 1365.036	39.04917 9.515636	49.09674 13.60235	118.7375 33.71433	4601.154 1552.966	2.343329 1.325529	44.62499 21.39307	82.93272 38.62163	83.67417 22.40809	6.4537 4.1591

-> Group = E, CavalOcclusion = During Occlusion

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	EDP
0	1994.018 754.22	24.83805 5.13788	38.48208 11.99936	85.97261 22.15682	2082.852 444.3339	2.891254 .8350047	42.29728 17.85128	64.74662 11.50167	67.13533 18.79899	4.5121 3.6992
30	1236.393 449.0936	29.54429 5.422712	55.21222 9.995526	133.7133 29.42764	3904.939 824.3368	1.477359 .304272	24.2122 7.730587	41.98038 7.267783	53.75649 7.755614	3.5515 4.342
74	3017.477 1454.442	27.9661 4.705415	41.11593 8.276152	130.2219 30.42981	3619.531 898.2643	3.643069 .8027798	41.67944 11.0684	99.29644 25.10739	69.64554 9.329961	5.083 3.268
Total	1961.739 1132.744	27.74874 5.159508	46.40467 12.29566	119.0756 33.67994	3302.797 1072.035	2.500103 1.130061	34.37 14.42889	64.86104 28.33386	62.11874 13.49911	4.263 3.652

-> Group = E, CavalOcclusion = Post Occlusion

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	EDP
0	3132.269 792.2669	42.69601 15.14697	44.93208 13.74686	85.50049 13.39493	3718.828 1547.128	2.001305 .804056	54.19037 22.0256	73.44024 7.503975	96.88638 22.38206	6.8834 3.5296
30	2530.218 641.389	43.61787 9.283529	64.26386 6.951952	118.5622 21.81629	5229.34 1841.108	1.288054 .3598582	24.78277 8.081213	53.48515 11.16984	68.40064 14.76964	2.5495 3.4793
74	4621.405 871.9186	34.24653 4.408769	40.82608 7.6661	135.3786 26.39738	4610.863 1042.846	3.958645 1.271488	51.94487 16.41099	123.7522 21.43894	86.1914 16.43345	5.954 4.5654
Total	3357.762 1135.625	40.53623 11.00736	50.54742 14.16899	111.8394 28.79697	4514.313 1584.643	2.325258 1.386789	43.15078 20.87413	81.1949 32.37214	83.68701 21.14602	5.0805 4.0922

6 .

7 .

8 . clear all

9 . import excel "C:\Users\fahim\Downloads\PV Loop Statistics-Raw Data.xlsx", sheet("FINAL-PreIVCO") firstrow
(24 vars, 54 obs)

10 . destring StrokeWork StrokeVolume EjectionFraction HeartRate CardiacOutput Elastance ESV ESP EDV EDP, replace for
StrokeWork already numeric; no replace
StrokeVolume already numeric; no replace
EjectionFraction already numeric; no replace
HeartRate already numeric; no replace
CardiacOutput already numeric; no replace
Elastance already numeric; no replace
ESV already numeric; no replace
ESP already numeric; no replace
EDV already numeric; no replace
EDP already numeric; no replace

11 . by TimeNumerical, sort : tabstat StrokeWork StrokeVolume EjectionFraction HeartRate CardiacOutput Elastance ESV

-> TimeNumerical = 0

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
0	3383.356 825.0453	45.944 15.27948	42.83344 10.59821	89.83882 16.748	4208.442 1899.564	1.81991 .6059787	62.30243 18.15292	74.37041 5.563195	108.2464 22.19321	6.7754 2.9286
Total	3383.356 825.0453	45.944 15.27948	42.83344 10.59821	89.83882 16.748	4208.442 1899.564	1.81991 .6059787	62.30243 18.15292	74.37041 5.563195	108.2464 22.19321	6.7754 2.9286

-> TimeNumerical = 30

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
30	1698.251 501.4066	41.0719 10.42821	60.48227 8.635566	127.4071 29.4324	5336.151 2104.659	1.088787 .3524749	27.21766 8.541832	41.21716 8.059421	68.28956 15.14512	4.0412 3.4171
Total	1698.251 501.4066	41.0719 10.42821	60.48227 8.635566	127.4071 29.4324	5336.151 2104.659	1.088787 .3524749	27.21766 8.541832	41.21716 8.059421	68.28956 15.14512	4.0412 3.4171

-> TimeNumerical = 74

Summary statistics: Mean, SD

Group variable: TimeNumerical (Time (Numerical))

TimeNumerical	Stroke~k	Stroke~e	Ejecti~n	HeartR~e	Cardia~t	Elasta~e	ESV	ESP	EDV	E
74	6187.183 3071.403	50.72679 22.25567	46.37994 10.69733	145.5684 24.20181	7482.653 3956.52	3.042104 1.765285	57.75666 17.12941	126.1774 48.00728	108.4835 30.58709	7.0067 2.5234
Total	6187.183 3071.403	50.72679 22.25567	46.37994 10.69733	145.5684 24.20181	7482.653 3956.52	3.042104 1.765285	57.75666 17.12941	126.1774 48.00728	108.4835 30.58709	7.0067 2.5234

12 . clear all

13 .

14 .

```

15 . import excel "C:\Users\fahim\Downloads\PV Loop Statistics-Raw Data.xlsx", sheet("FINAL-PreIVCO") firstrow
    (24 vars, 54 obs)

16 . destring StrokeWork StrokeVolume EjectionFraction HeartRate CardiacOutput Elastance ESV ESP EDV EDP, replace fo
    StrokeWork already numeric; no replace
    StrokeVolume already numeric; no replace
    EjectionFraction already numeric; no replace
    HeartRate already numeric; no replace
    CardiacOutput already numeric; no replace
    Elastance already numeric; no replace
    ESV already numeric; no replace
    ESP already numeric; no replace
    EDV already numeric; no replace
    EDP already numeric; no replace

17 .
18 .
19 .
20 . /* ESP -with interaction */
21 . destring ESP, replace force
    ESP already numeric; no replace

22 .
23 . xtset Number TimeNumerical

    Panel variable: Number (strongly balanced)
    Time variable: TimeNumerical, 0 to 74, but with gaps
    Delta: 1 unit

24 . xtmixed ESP GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
    Note: time gaps exist in the estimation data

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0:  log restricted-likelihood = -172.61687 (not concave)
Iteration 1:  log restricted-likelihood = -172.19927 (not concave)
Iteration 2:  log restricted-likelihood = -172.19896 (not concave)
Iteration 3:  log restricted-likelihood = -172.19894
Iteration 4:  log restricted-likelihood = -172.19894
Iteration 5:  log restricted-likelihood = -172.19894 (not concave)
Iteration 6:  log restricted-likelihood = -172.19848 (not concave)
Iteration 7:  log restricted-likelihood = -172.10416
Iteration 8:  log restricted-likelihood = -172.09818
Iteration 9:  log restricted-likelihood = -172.09817

Computing standard errors:

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =          2.9
    max =          3
Wald chi2(10)      =     686.20
Prob > chi2        =     0.0000
Log restricted-likelihood = -172.09817

```

ESP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	-3.838006	6.869675	-0.56	0.576	-17.30232	9.626309
2	3.501585	6.954977	0.50	0.615	-10.12992	17.13309
TimeNumerical						
30	-37.57618	6.449778	-5.83	0.000	-50.21752	-24.93485
74	-3.356995	6.834324	-0.49	0.623	-16.75202	10.03803
GroupNumerical#TimeNumerical						
1 30	5.229546	9.374675	0.56	0.577	-13.14448	23.60357
1 74	105.0288	9.665193	10.87	0.000	86.08532	123.9722
2 30	8.162301	9.121363	0.89	0.371	-9.715242	26.03984
2 74	62.83358	9.919139	6.33	0.000	43.39243	82.27474
Sex	2.312136	3.822856	0.60	0.545	-5.180524	9.804796
weight	-.6634233	.3434064	-1.93	0.053	-1.336487	.0096409
_cons	126.5646	27.19787	4.65	0.000	73.25771	179.8714

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	2.30e-17	.	.	.
Residual: AR(1)				
rho	.9300434	.0737827	.528217	.9915248
var(e)	140.7813	31.534	90.75735	218.3775

LR test vs. linear model: $\chi^2(2) = 0.20$ Prob > $\chi^2 = 0.9041$

Note: LR test is conservative and provided only for reference.

25 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
ESP			
TimeNumerical@GroupNumerical			
0	2	41.35	0.0000
1	2	395.62	0.0000
2	2	159.22	0.0000
Joint	6	597.12	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
ESP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-37.57618	6.449778	-5.83	0.000	-50.21752	-24.93485
(30 vs base) 1	-32.34664	6.803301	-4.75	0.000	-45.68086	-19.01241
(30 vs base) 2	-29.41388	6.449778	-4.56	0.000	-42.05522	-16.77255
(74 vs base) 0	-3.356995	6.834324	-0.49	0.623	-16.75202	10.03803
(74 vs base) 1	101.6718	6.834324	14.88	0.000	88.27673	115.0668
(74 vs base) 2	59.47659	7.188973	8.27	0.000	45.38646	73.56672

26 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
ESP			
GroupNumerical@TimeNumerical			
0	2	1.08	0.5822
30	2	3.24	0.1983
74	2	224.30	0.0000
Joint	6	228.30	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
ESP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-3.838006	6.869675	-0.56	0.576	-17.30232	9.626309
(1 vs base) 30	1.39154	7.192828	0.19	0.847	-12.70614	15.48922
(1 vs base) 74	101.1907	6.869675	14.73	0.000	87.72643	114.6551
(2 vs base) 0	3.501585	6.954977	0.50	0.615	-10.12992	17.13309
(2 vs base) 30	11.66389	6.954977	1.68	0.094	-1.967618	25.29539
(2 vs base) 74	66.33517	7.38304	8.98	0.000	51.86468	80.80566

27 .
 28 . /* Repeat with natural log */
 29 . generate logESP = ln(ESP) /*natural log transform*/
 (2 missing values generated)
 30 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)
 Time variable: **TimeNumerical**, 0 to 74, but with gaps
 Delta: 1 unit

31 . xtmixed logESP GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
 Note: time gaps exist in the estimation data

Obtaining starting values by EM:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **6.4884963** (not concave)
 Iteration 1: log restricted-likelihood = **6.4959048** (backed up)
 Iteration 2: log restricted-likelihood = **6.6572445** (not concave)
 Iteration 3: log restricted-likelihood = **6.9119543**
 Iteration 4: log restricted-likelihood = **6.9122545**
 Iteration 5: log restricted-likelihood = **6.9122561**
 Iteration 6: log restricted-likelihood = **6.9122561**

Computing standard errors:

Mixed-effects REML regression
 Group variable: **Number**

Number of obs	=	52
Number of groups	=	18
Obs per group:		
min	=	2
avg	=	2.9
max	=	3
Wald chi2(10)	=	555.35
Prob > chi2	=	0.0000

Log restricted-likelihood = **6.9122561**

	logESP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical							
1		-.0476986	.086846	-0.55	0.583	-.2179136	.1225163
2		.0465622	.0878398	0.53	0.596	-.1256006	.2187251
TimeNumerical							
30		-.7081317	.0857133	-8.26	0.000	-.8761267	-.5401367
74		-.0670574	.0866177	-0.77	0.439	-.2368249	.1027102
GroupNumerical#TimeNumerical							
1 30		.0857757	.1243199	0.69	0.490	-.1578869	.3294382
1 74		.9432333	.1224959	7.70	0.000	.7031458	1.183321
2 30		.2133294	.1212169	1.76	0.078	-.0242514	.4509101
2 74		.6345665	.125704	5.05	0.000	.3881912	.8809418
Sex		.0967339	.0463972	2.08	0.037	.005797	.1876709
weight		-.0081422	.0041667	-1.95	0.051	-.0163087	.0000243
_cons		4.92314	.3304659	14.90	0.000	4.275439	5.570842

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	6.65e-25	.	.	.
Residual: AR(1)				
rho	.8789525	.2294546	-.5414182	.9975336
var(e)	.0225095	.0049755	.0145954	.0347148

LR test vs. linear model: chi2(2) = **0.02** Prob > chi2 = **0.9896**

Note: LR test is conservative and provided only for reference.

32 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logESP			
TimeNumerical@GroupNumerical			
0	2	82.95	0.0000
1	2	278.32	0.0000
2	2	136.50	0.0000
Joint	6	498.63	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logESP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.7081317	.0857133	-8.26	0.000	-.8761267	-.5401367
(30 vs base) 1	-.622356	.0900482	-6.91	0.000	-.7988472	-.4458648
(30 vs base) 2	-.4948023	.0857133	-5.77	0.000	-.6627973	-.3268073
(74 vs base) 0	-.0670574	.0866177	-0.77	0.439	-.2368249	.1027102
(74 vs base) 1	.876176	.0866177	10.12	0.000	.7064084	1.045944
(74 vs base) 2	.5675091	.0910981	6.23	0.000	.38896	.7460582

33 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logESP			
GroupNumerical@TimeNumerical			
0	2	1.12	0.5712
30	2	9.89	0.0071
74	2	115.40	0.0000
Joint	6	125.83	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logESP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	-.0476986	.086846	-0.55	0.583	-.2179136	.1225163
(1 vs base) 30	.0380771	.0909954	0.42	0.676	-.1402706	.2164247
(1 vs base) 74	.8955347	.086846	10.31	0.000	.7253198	1.06575
(2 vs base) 0	.0465622	.0878398	0.53	0.596	-.1256006	.2187251
(2 vs base) 30	.2598916	.0878398	2.96	0.003	.0877287	.4320544
(2 vs base) 74	.6811287	.0932218	7.31	0.000	.4984174	.8638401

```

34 .
35 .
36 . /*ESP -without interaction */
37 . xtset Number TimeNumerical

```

Panel variable: **Number** (strongly balanced)
 Time variable: **TimeNumerical**, 0 to 74, but with gaps
 Delta: 1 unit

```
38 . xtmixed ESP Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml
```

Performing EM optimization:

Performing gradient-based optimization:

```

Iteration 0: log restricted-likelihood = -219.42788
Iteration 1: log restricted-likelihood = -218.96101
Iteration 2: log restricted-likelihood = -218.94663
Iteration 3: log restricted-likelihood = -218.94663

```

Computing standard errors:

```

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =         2.9
    max =          3
Wald chi2(6)       =     124.85
Prob > chi2        =     0.0000
Log restricted-likelihood = -218.94663

```

ESP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	2.237036	7.472358	0.30	0.765	-12.40852	16.88259
weight	-.6398394	.669447	-0.96	0.339	-1.951931	.6722526
GroupNumerical						
1	33.88068	8.296236	4.08	0.000	17.62036	50.14101
2	26.85233	8.658753	3.10	0.002	9.881485	43.82317
TimeNumerical						
30	-32.18169	8.246217	-3.90	0.000	-48.34398	-16.0194
74	52.36511	8.251154	6.35	0.000	36.19314	68.53707
_cons	104.3543	52.56394	1.99	0.047	1.330835	207.3777

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	9.41e-19	7.10e-18	3.56e-25	2.49e-12
var(Residual)	593.5681	125.1456	392.6508	897.2935

LR test vs. linear model: chibar2(01) = 0.00 Prob >= chibar2 = 1.0000

39 .

40 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
ESP			
TimeNumerical@GroupNumerical			
0	2	103.94	0.0000
1	2	103.94	0.0000
2	2	103.94	0.0000
Joint	2	103.94	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
ESP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-32.18169	8.246217	-3.90	0.000	-48.34398	-16.0194
(30 vs base) 1	-32.18169	8.246217	-3.90	0.000	-48.34398	-16.0194
(30 vs base) 2	-32.18169	8.246217	-3.90	0.000	-48.34398	-16.0194
(74 vs base) 0	52.36511	8.251154	6.35	0.000	36.19314	68.53707
(74 vs base) 1	52.36511	8.251154	6.35	0.000	36.19314	68.53707
(74 vs base) 2	52.36511	8.251154	6.35	0.000	36.19314	68.53707

41 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
ESP			
GroupNumerical@TimeNumerical			
0	2	18.95	0.0001
30	2	18.95	0.0001
74	2	18.95	0.0001
Joint	2	18.95	0.0001

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
ESP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	33.88068	8.296236	4.08	0.000	17.62036	50.14101
(1 vs base) 30	33.88068	8.296236	4.08	0.000	17.62036	50.14101
(1 vs base) 74	33.88068	8.296236	4.08	0.000	17.62036	50.14101
(2 vs base) 0	26.85233	8.658753	3.10	0.002	9.881485	43.82317
(2 vs base) 30	26.85233	8.658753	3.10	0.002	9.881485	43.82317
(2 vs base) 74	26.85233	8.658753	3.10	0.002	9.881485	43.82317

```

42 .
43 . /* Repeat with natural log */
44 . xtset Number TimeNumerical

Panel variable: Number (strongly balanced)
Time variable: TimeNumerical, 0 to 74, but with gaps
Delta: 1 unit

45 . xtmixed logESP Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

```

Performing EM optimization:

Performing gradient-based optimization:

```

Iteration 0: log restricted-likelihood = -11.609387
Iteration 1: log restricted-likelihood = -11.13775
Iteration 2: log restricted-likelihood = -11.135874
Iteration 3: log restricted-likelihood = -11.135874

```

Computing standard errors:

```

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =         2.9
    max =          3
Wald chi2(6)       =     186.22
Prob > chi2        =     0.0000

Log restricted-likelihood = -11.135874

```

logESP	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	.0941258	.0737698	1.28	0.202	-.0504603	.2387119
weight	-.0081365	.006609	-1.23	0.218	-.02109	.0048169
GroupNumerical						
1	.304457	.0819034	3.72	0.000	.1439293	.4649847
2	.3269132	.0854823	3.82	0.000	.159371	.4944554
TimeNumerical						
30	-.5992859	.0814096	-7.36	0.000	-.7588457	-.439726
74	.4566147	.0814583	5.61	0.000	.2969593	.6162701
_cons	4.712724	.5189298	9.08	0.000	3.69564	5.729807

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	2.05e-24	1.49e-23	1.35e-30	3.11e-18
var(Residual)	.0578512	.0121971	.0382691	.0874532

LR test vs. linear model: chibar2(01) = 0.00 Prob >= chibar2 = 1.0000

46 .

47 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logESP			
TimeNumerical@GroupNumerical			
0	2	164.20	0.0000
1	2	164.20	0.0000
2	2	164.20	0.0000
Joint	2	164.20	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logESP						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.5992859	.0814096	-7.36	0.000	-.7588457	-.439726
(30 vs base) 1	-.5992859	.0814096	-7.36	0.000	-.7588457	-.439726
(30 vs base) 2	-.5992859	.0814096	-7.36	0.000	-.7588457	-.439726
(74 vs base) 0	.4566147	.0814583	5.61	0.000	.2969593	.6162701
(74 vs base) 1	.4566147	.0814583	5.61	0.000	.2969593	.6162701
(74 vs base) 2	.4566147	.0814583	5.61	0.000	.2969593	.6162701

48 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logESP			
GroupNumerical@TimeNumerical			
0	2	19.96	0.0000
30	2	19.96	0.0000
74	2	19.96	0.0000
Joint	2	19.96	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logESP						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.304457	.0819034	3.72	0.000	.1439293	.4649847
(1 vs base) 30	.304457	.0819034	3.72	0.000	.1439293	.4649847
(1 vs base) 74	.304457	.0819034	3.72	0.000	.1439293	.4649847
(2 vs base) 0	.3269132	.0854823	3.82	0.000	.159371	.4944554
(2 vs base) 30	.3269132	.0854823	3.82	0.000	.159371	.4944554
(2 vs base) 74	.3269132	.0854823	3.82	0.000	.159371	.4944554

```

49 .
50 .
51 . /* ESV -with interaction */
52 . destring ESV, replace force
    ESV already numeric; no replace

```

```

53 .
54 . xtset Number TimeNumerical

```

Panel variable: **Number** (strongly balanced)
 Time variable: **TimeNumerical**, 0 to 74, but with gaps
 Delta: 1 unit

```

55 . xtmixed ESV GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
    Note: time gaps exist in the estimation data

```

Obtaining starting values by EM:

Performing gradient-based optimization:

```

Iteration 0:  log restricted-likelihood = -175.74066 (not concave)
Iteration 1:  log restricted-likelihood = -175.74066 (not concave)
Iteration 2:  log restricted-likelihood = -175.57864 (backed up)
Iteration 3:  log restricted-likelihood = -175.34961
Iteration 4:  log restricted-likelihood = -175.3175
Iteration 5:  log restricted-likelihood = -175.31725
Iteration 6:  log restricted-likelihood = -175.31725

```

Computing standard errors:

```

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =          2.9
    max =          3
Wald chi2(10)      =     169.84
Prob > chi2        =      0.0000

Log restricted-likelihood = -175.31725

```

ESV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	5.259623	9.261074	0.57	0.570	-12.89175	23.41099
2	-8.366745	9.48358	-0.88	0.378	-26.95422	10.22073
TimeNumerical						
30	-36.68464	5.066158	-7.24	0.000	-46.61412	-26.75515
74	-14.86648	6.202003	-2.40	0.017	-27.02219	-2.71078
GroupNumerical#TimeNumerical						
1 30	-.2311876	7.413888	-0.03	0.975	-14.76214	14.29977
1 74	15.36011	8.770957	1.75	0.080	-1.830647	32.55087
2 30	6.220737	7.164629	0.87	0.385	-7.821678	20.26315
2 74	16.81974	9.054704	1.86	0.063	-.9271497	34.56664
Sex	-2.442124	7.296598	-0.33	0.738	-16.74319	11.85895
weight	.1032668	.6458172	0.16	0.873	-1.162512	1.369045
_cons	55.92526	50.70582	1.10	0.270	-43.45633	155.3068

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	125.2623	85.24828	33.00106	475.4585
Residual: AR(1)				
rho	-.970253	.0317224	-.9963881	-.7762074
var(e)	129.2249	69.19051	45.24702	369.0648

LR test vs. linear model: $\chi^2(2) = 14.34$ Prob > $\chi^2 = 0.0008$

Note: LR test is conservative and provided only for reference.

56 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
ESV			
TimeNumerical@GroupNumerical			
0	2	54.90	0.0000
1	2	61.39	0.0000
2	2	49.70	0.0000
Joint	6	166.03	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
ESV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-36.68464	5.066158	-7.24	0.000	-46.61412	-26.75515
(30 vs base) 1	-36.91582	5.412927	-6.82	0.000	-47.52497	-26.30668
(30 vs base) 2	-30.4639	5.066158	-6.01	0.000	-40.39339	-20.53441
(74 vs base) 0	-14.86648	6.202003	-2.40	0.017	-27.02219	-2.71078
(74 vs base) 1	.4936301	6.202003	0.08	0.937	-11.66207	12.64933
(74 vs base) 2	1.953262	6.597183	0.30	0.767	-10.97698	14.8835

57 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
ESV			
GroupNumerical@TimeNumerical			
0	2	1.97	0.3727
30	2	0.55	0.7577
74	2	4.97	0.0831
Joint	6	8.32	0.2158

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
ESV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	5.259623	9.261074	0.57	0.570	-12.89175	23.41099
(1 vs base) 30	5.028435	9.450882	0.53	0.595	-13.49495	23.55182
(1 vs base) 74	20.61974	9.261074	2.23	0.026	2.468364	38.77111
(2 vs base) 0	-8.366745	9.48358	-0.88	0.378	-26.95422	10.22073
(2 vs base) 30	-2.146009	9.48358	-0.23	0.821	-20.73348	16.44147
(2 vs base) 74	8.453	9.813737	0.86	0.389	-10.78157	27.68757

```

58 .
59 . /* Repeat with natural log */
60 . generate logESV = ln(ESV) /*natural log transform*/
    (2 missing values generated)

61 . xtset Number TimeNumerical

Panel variable: Number (strongly balanced)
Time variable: TimeNumerical, 0 to 74, but with gaps
Delta: 1 unit

62 . xtmixed logESV GroupNumerical##TimeNumerical Sex weight || Number:, var residuals(ar, t(TimeNumerical)) reml
Note: time gaps exist in the estimation data

```

Obtaining starting values by EM:

Performing gradient-based optimization:

```

Iteration 0: log restricted-likelihood = -10.804789 (not concave)
Iteration 1: log restricted-likelihood = -10.804789 (not concave)
Iteration 2: log restricted-likelihood = -10.568065 (backed up)
Iteration 3: log restricted-likelihood = -8.9627993
Iteration 4: log restricted-likelihood = -8.4841515
Iteration 5: log restricted-likelihood = -8.3902357
Iteration 6: log restricted-likelihood = -8.363383
Iteration 7: log restricted-likelihood = -8.356638
Iteration 8: log restricted-likelihood = -8.3563074
Iteration 9: log restricted-likelihood = -8.3563073

```

Computing standard errors:

```

Mixed-effects REML regression
Group variable: Number
Number of obs      =      52
Number of groups   =      18
Obs per group:
    min =          2
    avg =          2.9
    max =          3
Wald chi2(10)      =     458.64
Prob > chi2        =     0.0000

Log restricted-likelihood = -8.3563073

```

logESV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
GroupNumerical						
1	.0889117	.1716486	0.52	0.604	-.2475134	.4253368
2	-.1644257	.1759582	-0.93	0.350	-.5092975	.1804461
TimeNumerical						
30	-.8599848	.0776059	-11.08	0.000	-1.01209	-.7078799
74	-.2435712	.1125958	-2.16	0.031	-.464255	-.0228874
GroupNumerical#TimeNumerical						
1 30	.0495736	.1132585	0.44	0.662	-.1724091	.2715562
1 74	.2270933	.1592346	1.43	0.154	-.0850007	.5391873
2 30	.0666142	.1097514	0.61	0.544	-.1484945	.281723
2 74	.3356047	.1637823	2.05	0.040	.0145973	.6566121
Sex	-.0423061	.1381861	-0.31	0.759	-.3131458	.2285335
weight	.0012299	.0122395	0.10	0.920	-.0227591	.0252188
_cons	4.033757	.9606215	4.20	0.000	2.150974	5.916541

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	8.93e-15	.	.	.
Residual: AR(1)				
rho	-.992308	.0033564	-.996733	-.9819434
var(e)	.0873786	.0275352	.0471162	.1620465

LR test vs. linear model: $\chi^2(2) = 23.75$ Prob > $\chi^2 = 0.0000$

Note: LR test is conservative and provided only for reference.

63 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logESV			
TimeNumerical@GroupNumerical			
0	2	152.15	0.0000
1	2	137.21	0.0000
2	2	165.60	0.0000
Joint	6	455.01	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logESV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.8599848	.0776059	-11.08	0.000	-1.01209	-.7078799
(30 vs base) 1	-.8104112	.0824913	-9.82	0.000	-.9720912	-.6487313
(30 vs base) 2	-.7933706	.0776059	-10.22	0.000	-.9454754	-.6412657
(74 vs base) 0	-.2435712	.1125958	-2.16	0.031	-.464255	-.0228874
(74 vs base) 1	-.0164779	.1125958	-0.15	0.884	-.2371617	.2042059
(74 vs base) 2	.0920336	.1189404	0.77	0.439	-.1410853	.3251525

64 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logESV			
GroupNumerical@TimeNumerical			
0	2	2.00	0.3681
30	2	1.68	0.4308
74	2	3.41	0.1816
Joint	6	7.25	0.2982

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logESV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.0889117	.1716486	0.52	0.604	-.2475134	.4253368
(1 vs base) 30	.1384853	.1738895	0.80	0.426	-.2023319	.4793025
(1 vs base) 74	.316005	.1716486	1.84	0.066	-.0204201	.6524301
(2 vs base) 0	-.1644257	.1759582	-0.93	0.350	-.5092975	.1804461
(2 vs base) 30	-.0978115	.1759582	-0.56	0.578	-.4426833	.2470603
(2 vs base) 74	.171179	.1813015	0.94	0.345	-.1841654	.5265234

65 .
66 .
67 . /*ESV -without interaction */
68 . xtset Number TimeNumerical

Panel variable: **Number** (strongly balanced)
Time variable: **TimeNumerical**, 0 to 74, but with gaps
Delta: 1 unit

69 . xtmixed ESV Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-190.4782**

Iteration 1: log restricted-likelihood = **-190.4782**

Computing standard errors:

Mixed-effects REML regression

Group variable: **Number**

Number of obs = **52**

Number of groups = **18**

Obs per group:

min = **2**

avg = **2.9**

max = **3**

Wald chi2(6) = **115.08**

Prob > chi2 = **0.0000**

Log restricted-likelihood = **-190.4782**

ESV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-2.515209	7.007816	-0.36	0.720	-16.25027	11.21986
weight	.0638142	.6184842	0.10	0.918	-1.148393	1.276021
GroupNumerical						
1	10.52432	7.719342	1.36	0.173	-4.605313	25.65395
2	-.7262781	7.98606	-0.09	0.928	-16.37867	14.92611
TimeNumerical						
30	-34.40722	3.517716	-9.78	0.000	-41.30182	-27.51262
74	-4.317641	3.518425	-1.23	0.220	-11.21363	2.578345
_cons	54.79095	48.46483	1.13	0.258	-40.19837	149.7803

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	139.178	69.50379	52.29873	370.3823
var(Residual)	107.0188	26.68652	65.64506	174.469

LR test vs. linear model: chibar2(01) = 13.18 Prob >= chibar2 = **0.0001**

70 .

71 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
ESV			
TimeNumerical@GroupNumerical			
0	2	111.25	0.0000
1	2	111.25	0.0000
2	2	111.25	0.0000
Joint	2	111.25	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
ESV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-34.40722	3.517716	-9.78	0.000	-41.30182	-27.51262
(30 vs base) 1	-34.40722	3.517716	-9.78	0.000	-41.30182	-27.51262
(30 vs base) 2	-34.40722	3.517716	-9.78	0.000	-41.30182	-27.51262
(74 vs base) 0	-4.317641	3.518425	-1.23	0.220	-11.21363	2.578345
(74 vs base) 1	-4.317641	3.518425	-1.23	0.220	-11.21363	2.578345
(74 vs base) 2	-4.317641	3.518425	-1.23	0.220	-11.21363	2.578345

72 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
ESV			
GroupNumerical@TimeNumerical			
0	2	2.45	0.2938
30	2	2.45	0.2938
74	2	2.45	0.2938
Joint	2	2.45	0.2938

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
ESV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	10.52432	7.719342	1.36	0.173	-4.605313	25.65395
(1 vs base) 30	10.52432	7.719342	1.36	0.173	-4.605313	25.65395
(1 vs base) 74	10.52432	7.719342	1.36	0.173	-4.605313	25.65395
(2 vs base) 0	-.7262781	7.98606	-0.09	0.928	-16.37867	14.92611
(2 vs base) 30	-.7262781	7.98606	-0.09	0.928	-16.37867	14.92611
(2 vs base) 74	-.7262781	7.98606	-0.09	0.928	-16.37867	14.92611

```

73 .
74 . /* Repeat with natural log */
75 . xtset Number TimeNumerical

Panel variable: Number (strongly balanced)
Time variable: TimeNumerical, 0 to 74, but with gaps
Delta: 1 unit

76 . xtmixed logESV Sex weight i.GroupNumerical i.TimeNumerical || Number:, var reml

```

Performing EM optimization:

Performing gradient-based optimization:

Iteration 0: log restricted-likelihood = **-9.2254986**

Iteration 1: log restricted-likelihood = **-9.2254986**

Computing standard errors:

Mixed-effects REML regression	Number of obs	=	52
Group variable: Number	Number of groups	=	18
	Obs per group:		
	min	=	2
	avg	=	2.9
	max	=	3
	Wald chi2(6)	=	229.51
Log restricted-likelihood = -9.2254986	Prob > chi2	=	0.0000

logESV	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
Sex	-.0433368	.1392148	-0.31	0.756	-.3161927	.2295192
weight	-.0006125	.0122704	-0.05	0.960	-.024662	.023437
GroupNumerical						
1	.1794659	.1532373	1.17	0.242	-.1208736	.4798054
2	-.0325931	.1584124	-0.21	0.837	-.3430757	.2778896
TimeNumerical						
30	-.8201601	.0600015	-13.67	0.000	-.9377609	-.7025594
74	-.0646937	.0600107	-1.08	0.281	-.1823126	.0529252
_cons	4.106746	.9613993	4.27	0.000	2.222438	5.991054

Random-effects parameters	Estimate	Std. err.	[95% conf. interval]	
Number: Identity				
var(_cons)	.058663	.0273756	.0235041	.1464149
var(Residual)	.03111	.0077699	.0190681	.0507567

LR test vs. linear model: chibar2(01) = 18.51 Prob >= chibar2 = **0.0000**

77 .

78 . contrast TimeNumerical@GroupNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logESV			
TimeNumerical@GroupNumerical			
0	2	225.68	0.0000
1	2	225.68	0.0000
2	2	225.68	0.0000
Joint	2	225.68	0.0000

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logESV						
TimeNumerical@GroupNumerical						
(30 vs base) 0	-.8201601	.0600015	-13.67	0.000	-.9377609	-.7025594
(30 vs base) 1	-.8201601	.0600015	-13.67	0.000	-.9377609	-.7025594
(30 vs base) 2	-.8201601	.0600015	-13.67	0.000	-.9377609	-.7025594
(74 vs base) 0	-.0646937	.0600107	-1.08	0.281	-.1823126	.0529252
(74 vs base) 1	-.0646937	.0600107	-1.08	0.281	-.1823126	.0529252
(74 vs base) 2	-.0646937	.0600107	-1.08	0.281	-.1823126	.0529252

79 . contrast GroupNumerical@TimeNumerical , effect

Contrasts of marginal linear predictions

Margins: **asbalanced**

	df	chi2	P>chi2
logESV			
GroupNumerical@TimeNumerical			
0	2	2.02	0.3644
30	2	2.02	0.3644
74	2	2.02	0.3644
Joint	2	2.02	0.3644

	Contrast	Std. err.	z	P> z	[95% conf. interval]	
logESV						
GroupNumerical@TimeNumerical						
(1 vs base) 0	.1794659	.1532373	1.17	0.242	-.1208736	.4798054
(1 vs base) 30	.1794659	.1532373	1.17	0.242	-.1208736	.4798054
(1 vs base) 74	.1794659	.1532373	1.17	0.242	-.1208736	.4798054
(2 vs base) 0	-.0325931	.1584124	-0.21	0.837	-.3430757	.2778896
(2 vs base) 30	-.0325931	.1584124	-0.21	0.837	-.3430757	.2778896
(2 vs base) 74	-.0325931	.1584124	-0.21	0.837	-.3430757	.2778896

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end of do-file

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